Marmara University Faculty of Engineering	**EE4047 MIDTERM EXAM**	**06.05.2023**

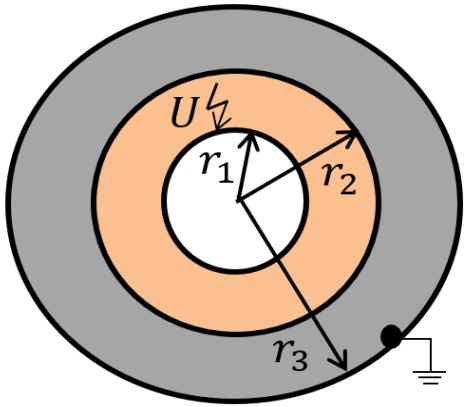
EXAM DURATION: **75 minutes**

NOTE: In all the questions, please consider **the last four digits of your student number** as **K, L, M** and **N** parameters. If **any number is 0**, use **the first non-zero number at the right of it**.
Example: 15062**3045** ➡ K=3, L=4, M=4, N=5

QUESTION-1 (30 pts): In the multilayer concentric spherical electrode system with two dielectric layers shown in the figure below, **U=140 kV**, r_1 **=0,6 cm**, r_2 **= 2,8 cm**,r_3 **= 3, 4 cm**, ε_{r1} **=** **K** and ε_{r2} **=** **L**.

a) **(15 pts)** Calculate the potential, minimum and maximum electric fields at each layer.

b) **(15 pts)** Considering breakdown strength for the layers as E_{b1} **=M×20 kV/cm** and E_{b2} **=N×35 kV/cm** find out if the system will withstand or not.



Equations:

$$A = \sum_{i=1}^{2} \frac{1}{\varepsilon_{ri}} \frac{r_{i+1} - r_i}{r_i . r_{i+1}}$$

$$U_i = \frac{U}{A . \varepsilon_{ri}} \frac{r_{i+1} - r_i}{r_i . r_{i+1}}$$

$$E_i = U_i \frac{r_i . r_{i+1}}{(r_{i+1} - r_i)} \frac{1}{r^2}$$

QUESTION-2 (25pts): Please,

a) Draw a single stage impulse generator circuit.

b) Explain **the role of each component** in the circuit.

c) Explain **how one impulse is generated** by the circuit, **step by step, in details**.

QUESTION-3 (30 pts): In a planar electrode system, electrode gap is **K *mm*,** electrode surface area is **15** cm^2, relative dielectric constant (relative permittivity) is **L**, applied voltage is **M× 30 kV,** dielectric breakdown strength of the insulation Material is **N× 15 kV/cm**.

Calculate: **(6 pts each)**

a) Electric field strength,

b) Capacity,

c) Charge,

d) Electric flux density,

e) The maximum voltage that can be applied.

Equations:

$$Q = C . U$$

$$D = \varepsilon . E$$

$$C = \frac{\varepsilon . S}{a}$$

$$E = \frac{U}{a}$$

$$\varepsilon_0 = 8{,}854 . 10^{-12}\ \text{F/m}$$

QUESTION-4 (15 pts): Please choose and answer **ONLY ONE OF THE FOLLOWING**:

Option 1: Draw and explain in details how high voltage DC is generated using **electrostatic generator**.

Option 2: Draw and explain in details how high voltage AC is generated using **two stage cascaded transformers.**